

# Osaid Khan

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## EDUCATION

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| <b>Pace University</b><br><i>Master of Science (MS) in Computer Science (GPA - 4.0/4.0)</i>                      | New York City, NY<br><i>Jan 2024 - Dec 2025</i> |
| <b>Dr. A.P.J. Abdul Kalam Technical University</b><br><i>Bachelor of Technology (B.Tech) in Computer Science</i> | Noida, India<br><i>Jul 2016 - Sep 2020</i>      |

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript/TypeScript, Rust  
**AI & Frameworks:** LangGraph, LangChain, RAG, LangSmith, FastMCP, PyTorch, FastAPI, Spring Boot  
**Databases & Messaging:** Vector Databases, PostgreSQL, Redis, Kafka, MongoDB  
**Integrations & Cloud:** SSO (SAML), SCIM, RESTful APIs, AWS (S3, Lambda, Step Functions), Google Cloud  
**AI Agents:** Claude code, Codex, Pi coding-agent, Agentic Workflows  
**Others:** Git, CI/CD, Webhooks, Reactive Programming, Node.js

## PROFESSIONAL EXPERIENCE

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| <b>Software Engineer</b><br>Sperse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Phoenix, AZ<br><i>Feb 2026 - present</i> |
| <ul style="list-style-type: none"><li>Architected and deployed an <b>Agentic AI</b> interaction layer using <b>LangChain</b> and <b>LangGraph</b> that translates natural-language intents into <b>safe, validated tool calls</b> across <b>1000+ platform endpoints</b></li><li>Built <b>agentic workflows</b> for automated customer actions, leveraging <b>FastAPI</b> and <b>FastMCP</b> to create high-velocity, reliable integration patterns between disparate systems and platform <b>APIs</b></li><li>Implemented a production-grade <b>observability layer</b> using <b>LangSmith</b> for <b>tracing, structured logs, and evaluation hooks</b> to monitor <b>prompt design</b> performance and iterate on <b>agentic reasoning</b></li><li>Optimized <b>prompt routing</b> and agent concurrency to maximize <b>KV-cache hit rate</b>, reducing end-to-end latency by <b>60%</b> and lowering LLM API costs by <b>50%</b> through efficient resource utilization</li><li>Developed <b>generative code solutions</b> (API-aware templates and SDK patterns) to accelerate <b>third-party integrations</b> and ensure seamless data flow across <b>distributed microservices</b></li></ul> |                                          |

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| <b>AI Researcher</b><br>Pace Artificial Intelligence Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | New York City, NY<br><i>Sep 2025 - Dec 2025</i> |
| <ul style="list-style-type: none"><li>Conducted research on <b>Fine-tuning</b> and <b>LORA (Low-rank Adaptation)</b> for <b>Stable Diffusion</b> and <b>StyleGAN</b> models, leveraging Pace University's <b>HPC cluster</b> for high-performance <b>model training</b></li><li>Utilized <b>vLLM</b> for model inference and <b>Hugging Face</b> for deployment, ensuring secure and efficient serving of custom-trained <b>LLMs</b> and generative models</li><li>Led workshops on building <b>AI Agents</b> with <b>LangGraph</b> and <b>LangSmith</b>, and GPU programming with <b>CUDA</b>, for <b>+50</b> students to promote <b>iterative refinement</b> in AI development</li></ul> |                                                 |

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| <b>Software Engineer</b><br>Qualitest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Noida, India<br><i>Feb 2021 - Dec 2023</i> |
| <ul style="list-style-type: none"><li>Built an internal <b>Agentic HR</b> support chatbot using <b>LangChain</b> and <b>RAG (Retrieval-Augmented Generation)</b> with a <b>PostgreSQL</b> vector store, synchronizing customer content for automated question-answering</li><li>Developed a cloud-agnostic Storage Service using <b>Python</b> and <b>Redis</b> that handles complex <b>data ingestion</b> and file operations across <b>S3</b> and <b>Google Cloud Storage</b> via <b>RESTful APIs</b> and signed URLs</li><li>Contributed to an Identity &amp; Access Management (<b>IAM</b>) platform supporting <b>SSO, SAML, and SCIM</b> protocols for thousands of users across 30+ applications using <b>React, Spring Boot, and AWS</b></li><li>Engineered backend workflows using <b>AWS Step Functions</b> to automate <b>data maintenance</b> and handle long-running <b>HRIS</b> data import jobs, preventing <b>AWS Lambda</b> timeouts</li></ul> |                                            |

## PROJECTS

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| <b>Dark Factory - Orchestrator for AI coding CLIs</b>   <i>Crosstern, Ratatui, Y-crdt</i>                                                                                                                    | <a href="https://github.com/ozzyozbourne/dark-factory">github.com/ozzyozbourne/dark-factory</a> |
| <ul style="list-style-type: none"><li>Orchestrates <b>agentic loops</b> via <b>PTY</b>, providing real-time event-driven updates and secure execution of <b>Python</b> scripts and system commands</li></ul> |                                                                                                 |

- Utilizes **Y-crdt** for distributed state synchronization across machines, enabling collaborative and reliable **AI-driven** automation workflows